



Improving the accuracy of collaborative filtering recommendations using clustering and association rules mining on implicit data



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ABSTRACT

The recommender systems are recently becoming more significant in the age of rapid development of the Internet technology due to their ability in making a decision to users on appropriate choices. Collaborative filtering (CF) is the most successful and most applied technique in the design of recommender systems where items to an active user will be recommended based on the past rating records from like-minded users. Unfortunately, CF may lead to the poor recommendation when user ratings on items are very sparse in comparison with the huge number of users and items in user-item matrix. To overcome this problem, this research applies the users' implicit interaction records with items to efficiently process massive data by employing association rules mining. It captures the multiple purchases per transaction in association rules, rather than just counting total purchases made. To do this, a modified preprocessing is implemented to discover similar interest patterns among users based on multiple purchases done. In addition, the clustering technique has been employed in our technique to reduce the size of data and dimensionality of the item space as the performance of association rules mining. Then, similarities between items based on their features were computed to make recommendations. The experiments were conducted and the results were compared with basic CF and other extended version of CF techniques including K-Means clustering, hybrid representation, and probabilistic learning by using public dataset, namely, Million Song dataset. The experimental results demonstrated that our technique achieves the better performance when compared to the basic CF and other extended version of CF techniques in terms of Precision, Recall metrics, even when the data is very sparse.

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1. Introduction

Due to the increased overloading problem of information, the recommender systems are becoming more significant in the age of rapid development of the Internet technology. Recommender systems have become an essential mechanism which provides users with useful selected information; in which this could be effective in making a decision for example in purchasing a product, selecting a movie to watch or doing any other online activity that requires making a choice or a decision (Huang, Ma, Cheng, & Wang, 2015).

The recommendation system can be employed online for certain types of business or activity, such as e-commerce (e.g., Amazon¹ a

book recommendation system), online news aggregators (e.g., Digg²) and online video sharing (e.g., YouTube³). This is because the Internet offers a wealth of information (such as online news, books, articles, music, movies and other products) that may be useful to potential purchasers or consumers (Kim, El-Saddik, & Jo, 2011a; Kim, Ha, Lee, Jo, & El-Saddik, 2011b).

Hence, a recommender system provides an alternative to discover items that users might not have found by themselves. One of the most successful technologies among the recommender systems is Collaborative Filtering (CF) which is based on explicit rating feedback on the items/products by like-minded users (known as neighbors) (Huang et al., 2015; Kim et al., 2011a,b).

Many online companies and commercial systems (such as movie

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¹ www.amazon.com.

² www.digg.com.

³ www.youtube.com.

recommendations in [Netflix.com](http://www.netflix.com),⁴ book recommendations in [Amazon.com](http://www.amazon.com), music recommendations in Last.fm⁵ and etc.) apply CF to generate recommendations to their customers. CF is the most applied and successful technique in a recommender system because it provides the best results and accurate recommendations even though it has simplistic algorithm (Kaleli, 2014; Kim et al., 2011a,b; Nakatsuji, Toda, Sawada, Zheng, & Hendler, 2016).

Unfortunately, CF leads to the poor recommendation when user ratings on items are very sparse in comparison with the huge number of users and items in user-item matrix (data sparsity problem). In the case of a lack of user rating on items, implicit data is used to profile a user's item preferences. Implicit data can indicate users' preferences by providing more evidences and information through observations made on users' behaviors. Furthermore, the data mining techniques such as clustering (Boratto, Carta, & Fenu, 2015; Zahra et al., 2015), classification (Kim et al., 2011b; Pan, Zhong, Xu, & Ming, 2015), Singular Value Decomposition (SVD) (Zhou, He, Huang, & Zhang, 2015), association rules mining (Kardan and Ebrahimi, 2013; Tyagi & Bharadwaj, 2013) have been applied to recommender systems as solutions to the data sparsity problem to provide possible connections between users and items based on retrieving either explicitly (user's ratings) or implicitly of user preferences and obtain the most efficient results (Xie, Chen, Shang, & Fox, 2014; Zhou et al., 2015). However, few research have considered developing recommender systems by association rules mining thus; there are still some issues on association rules mining that need to be addressed in the light of the emerging recommendation systems. In addition, most of clustering-based CF techniques only utilize historical rating information in clustering procedure but ignore other data resources in recommender systems such as social relationship interactions among users (tagging or user's listening behaviors) and correlations between items. Section 2 (Related work) provide more detail on different solutions to sparsity problem and analyze the research issues that are still unsolved or overlooked in the past studies.

To tackle data sparsity problem, this research integrates the correlations between items and the correlation between user's profile data into the framework of association rules mining and clustering technique. This research proposes an improved CF technique for personalized recommendation by exploring and exploiting the user profiles created from implicit user feedbacks. Clustering technique (hierarchical structure) can be viewed for efficiently profiling user's item preferences from play records of songs and as a data dimensionality reduction for the rules mining part. More notably, the proposed technique in this research focus on employing association rules mining technique (Apriori algorithm) with item repetition within a transaction (each user plays/listen a set of items frequently), which can increase the chance of improvement in CF. Then, this technique involves the calculated similarities between items based on their features to make recommendations.

It is important to know, none of previous research papers consider how to process massive data efficiently through employing association rules mining to model user's behaviors and how to use item repetition within a transaction (each user buys/uses a set of items) as input for association rules mining technique.

The major contributions of the study can be summarized as in the following paragraphs:

1. At first, the novelty of this research is to improve the accuracy of collaborative filtering recommendations in sparse data by

application of the clustering and association rules mining technique on implicit data. It is the first academic research to capture multiple purchases per transaction in association rules (counting problem), rather than just counting total purchases made. To do this, a modified preprocessing using the distribution of item repetition within a transaction as input for association rules mining are implemented to discover similar interest patterns among users. Application of association rules mining in this research will be explained in Sub-section 4.2.

2. Another uniqueness of this research relates to the processing the massive data efficiently through association rules mining to model user's behaviors. To achieve this capability, clustering technique can be viewed as a data dimensionality reduction for the rules mining part. This issue will be explained in detail in Sub-section 4.1.
3. We perform extensive experiments on real dataset, namely, Million Song dataset (MSD) in order to evaluate our proposed technique and compare it with basic CF. The results show that our proposed technique can indeed cope effectively and accurately with data sparsity and has higher accuracy than the basic technique in the recommendation process. Additionally, the experimental results shows the better performance when compared to the other extended version of CF techniques including K-Means clustering (Zahra et al., 2015), hybrid representation (Horsburgh, Craw, & Massie, 2015), and probabilistic learning (Lee, Kim, & Lee, 2010) in terms of Precision, Recall metrics, even when the data is sparse and few songs are played by each user.

It is interesting to note that most of the proposed recommender systems model the count data or user activity data logs (like viewing times, song listening count, and sales times of products) with simplifying them in binary data in terms of an arbitrarily chosen threshold whether users play the music or not, customer purchased a product or not (Horsburgh et al., 2015; Kolomvatsos, Anagnostopoulos, & Hadjiefthymiades, 2014; Nakatsuji et al., 2016). But this simplification might filter out a lot of suitable information. To stay away from this disadvantage, implicit feedback given by count data is all employed in our technique to use information on how often every user played each song. Thus, no information is lost.

The rest of this paper is structured as: Section 2 briefly discusses previous studies related to solving sparsity problem. In section 3, the theories related to the CF and association rules mining will be primarily introduced and then, in Section 4 the proposed technique is discussed. Section 5 explains the experimental methodology employed and the comparison made between the proposed technique and the primary CF. Finally, Section 6 provides the conclusions and the future directions of our work.

2. Related work

In this section, a brief review of literature which is related to collaborative recommender systems in solving sparsity problem is presented. In these recent years, significant steps have been taken in the direction of providing personalized services for a wide variety of web-based applications such as recommending news (Li, Wang, Chen, & Lin, 2010), videos (Choi & Suh, 2013), e-business applications (Kim, Yum, Song, & Kim, 2005), books (Langseth & Nielsen, 2015), movies (Acilar & Arslan, 2009), resource recommendations in social annotation systems (Li et al., 2010), bundle purchases (Choi et al., 2012) and online research papers (Najafabadi & Mahrin, 2016). CF is among the best-known recommendation recommender systems technique for producing recommendations, and there are currently a number of popular online companies such

⁴ <http://www.netflix.com>.

⁵ <http://www.last.fm>.

as Netflix.com, Amazon.com, and Last.fm that use CF to suggest services, products, and information of items to potential consumers (Najafabadi & Mahrin, 2016).

CF technique can be classified into user-based and item-based. In the user-based CF, a user will receive recommendations based on interests of like-minded users. In the item-based CF, a user will receive recommendations based on comparing similarity between items by considering the users who have rated both the items in user/item matrix (Shinde & Kulkarni, 2012). CF technique builds a user-item rating matrix by explicitly asking a user to rate an item on a sliding scale. Then, CF use similarity measurement methods to compute similarity between users or items based on user's rating scores for making prediction on empty cells of matrix. In most recommender systems by increasing the number of items, each user is unable to tell about their preferences on all available items and cannot rate millions of them. Thus, most cells of the user-item matrix are empty. In such cases, determining similar users or items (neighborhood formation) becomes a challenge. This is because the similarity between two users or items cannot be calculated since there is not enough information about ratings of the users and thus, the recommendation accuracy becomes very low (Najafabadi & Mahrin, 2016).

To overcome this disadvantage, several researchers have developed the data mining algorithms for filtering unseen items or employing pure rating data to make predictions, such as clustering CF models, dimensionality reduction techniques, Bayesian belief nets (BNs) CF models, link analysis, pattern mining approach and latent semantic CF models (Najafabadi & Mahrin, 2016; Park, Kim, Choi, & Kim, 2012). In particular, one of solutions for solving data sparsity in CF is Singular Value Decomposition (SVD) method (Zhou et al., 2015). It is commonly used to reduce the dimensionality of user-item ratings matrix in CF techniques. SVD can reduce the space in a user-item matrix and increase ratings density and find more ratings by finding hidden relationships from user-item matrix. Zhou et al. (2015) proposes an incremental approach based on SVD that repeatedly compute singular value decomposition of original matrix unchanged each time to solve sparsity problem and users' interests that are dynamic (Zhou et al., 2015).

Link Analysis explores trends and pattern by building up networks of interconnected objects (users or items). Link analysis finds the associations between objects in a database to solve sparsity problem (Gharibshah & Jalili, 2014). It has presented the potentials in enhancing the performance of web search. Social network analysis is one type of link analysis technique to discover a fundamental social structure via analyzing the patterns relationships and interactions between social actors. Social network analysis provides recommendation by employing users' social interaction (such as making social comments, social tags, online friending, etc.). Link analysis can also employ graph structure to obtain information on relations between users and items. Most of link analysis algorithms utilize a single node in the web graph to handle a web page (Park et al., 2012).

Zahra et al. (2015) mentioned that clustering technique is used to decrease the dimensionality of sparse rating matrices. This technique is based on the notion that within a small sub-area, users tend to associate with each other better than within the whole domain. Due to the fact that clustered sub-matrices are likely more dense than an original large matrix, better correlations are expected to be found (Zahra et al., 2015). Zahra et al. (2015) improved recommendation process by exploiting users with maximum number of ratings for clustering user/item matrix and finding out the most similar centroids as neighbors of active user.

Beside the clustering technique, the association rules mining technique has also been applied to represent users' interests in various fields for providing recommendation models. This is due to

its ability to scale to large datasets and achieve high precision (Tyagi & Bharadwaj, 2013). Kardan and Ebrahimi (2013) employed the association rules mining technique in a hybrid recommendation system to compute similarity between users from implicit data collected in a discussion group, such as adding comments or creating posts (Kardan & Ebrahimi, 2013). Feng, Tian, Wang, and Li (2015) proposed a novel recommendation technique in which temporal factors of users' interest were incorporated in both overlapping community detection based on a user–user graph generation with time-weighted links and association rules generation based on temporal overlapping communities to model user interest drift over time (Feng et al., 2015). It is interesting to note, studies on CF research using association rules mining are still deficient. For example, association rules mining techniques that were employed by Feng et al. (2015), Kardan and Ebrahimi (2013) and Tyagi and Bharadwaj (2013) did not consider the counting problem (item repetition within a transaction or multiple purchases per transaction). In more previous researches, item repetition within a transaction are often simplified with a binary transformation of the data (the item is represented by 1 if it has been purchased/seen or not as 0) which is a typical technique with implicit data. So, association rules mining techniques have been run on binary data (purchased or not, viewed or not, played or not). But this simplification in a sparse condition which is the problem of this research may filter out a lot of useful information. Therefore, implementing the CF research using association rules mining will be an interesting area for further research. Furthermore, none of previous research papers consider how to process massive data efficiently through employing association rules mining to predict the user's behaviors in the future. In this research, we adapt clustering technique to process massive data efficiently through employing association rules mining for identifying similar listening history among users on the same set of songs and predicting users' unknown preferences.

According to Najafabadi and Mahrin (2016); Shi, Larson, & Hanjalic, 2014, in addition to traditional emphasis on applying algorithms for improving neighborhood formation phase in CF, exploiting the additional information sources beyond the user/item matrix has been an important consideration by researchers (Shi et al., 2014). In term of interaction between users and items, recommender systems rely on different types of input data for making recommendations. The most convenient interaction to be used is the high quality explicit feedback, which includes explicit input by users regarding their interest in products. For example, Netflix⁶ collects star ratings for movies and TiVo users indicate their preferences for TV shows by hitting thumbs up/down buttons. However, explicit feedback is not always available. Thus, recommenders can infer users' preferences from the abundant implicit feedback, which indirectly reflects opinion through observing users' behavior (Pan & Yang, 2013). In the case of implicit feedback, information observed implicitly from users' behaviors is treated as preference indicators, what has a user listened to, accessed, viewed, or bought (Nakatsuji and Fujiwara (2014); Pan and Yang (2013)). In this work, our dataset contains implicit information like music listening information and tagging activities of the users and song's features like title, artist, release, song_hotness, year, duration to derive users' interests on song's features.

In general, based on researches done by Nakatsuji and Fujiwara (2014), the auxiliary data of the form “like” or “dislike” may be more easily obtained and overcome data sparsity for CF in numerical ratings, such as the favored/disfavored data in Moviepilot⁷

⁶ <https://www.netflix.com/my/>.

⁷ <http://www.moviepilot.de>.

Table 1
Literature reviews on different recommendation techniques in solving sparsity problem.

	Recommendation techniques
Process	Data mining (Dimensionality reduction, association rules, web mining)
Advantages	Recommends items based on learning of patterns among user or matching of item attributes to the user profile
Disadvantages	No sparsity problem, Sensitive to change of user preferences, Useful for hybrid recommender systems, Simplify and summarize the data
Research gaps in past studies	Need to know how it work and can be fitted on inputs and datasets
Dimensionality reduction (Clustering, SVD) (Zhou et al., 2015; Zhou et al., 2015)	- Potentially useful information might be lost in reducing dimensionality of the user and product space - Most of clustering-based CF only utilize historical rating information in clustering procedure but ignore other data resources in recommender systems such as social connections of users and correlations between items)
Association rules mining (Kardan & Ebrahimi, 2013; Tyagi & Bharadwaj, 2013)	- No sensitive to changes of user preferences - Does not consider multiple purchases per transaction (the number of times a user has purchased or used a product) - Does not consider how to process massive data efficiently through employing association rules mining
Link Analysis (Gharibshah & Jalili, 2014)	- Consider just for correlations between items no user's profile data - Increase amount of time and space used in analyzing and processing data - Need to learning of complex patterns (discovery of sequential patterns of consumption exhibited by the entire set of users). Complexity is a measure of the amount of time and space used by an algorithm in terms of an input of a given size.
Process	Implicit Feedback
Advantages	Recommend items based on tracking user' behaviors
Disadvantages	Useful for hybrid recommender system
Research gaps in past studies	Need to analyze user' behaviors
(Cheng & Wang, 2014)	- Convert implicit data to binary scale (purchased or not) to find similarity between users and items, thus some information is lost - Few research grasp user' interests from tags and categories data which is still new. Thus, there are still deficiencies on using tags in studies.
Process	Item-based
Advantages	Finds similar item based on rating data and predicts unrated items using weighted average of similar items
Disadvantages	No content analysis and domain independent
Research gaps in past studies	Not very accurate recommendation, sometimes happen sparsity problem
(Shinde & Kulkarni, 2012)	- Finding the relations between items no users
Process	Hybrid recommendation
Advantages	Combines two or more techniques for making recommendation
Disadvantages	Try to cover the weakness of other approaches and increase recommendations' accuracy
	Need to know process of each technique and how they work with each other

and the love/ban data in Last.fm and the “Want to see”/“Not interested” data in Flixster.⁸ It is more convenient for users to express such preferences instead of numerical ratings. There are some prior researches on applying such auxiliary data, the numerical ratings and implicit data of “whether rated” (Shinde & Kulkarni, 2012) or “whether purchased” (Cheng & Wang, 2014) or “click stream data” (Choi et al., 2012) to help boost the recommendation accuracy. Cheng and Wang (2014) have also changed the implicit data of simulated purchases to a user-brand matrix as a user-side metadata representing brand loyalty and a user-item matrix of “purchased” (Cheng & Wang, 2014).

This section provided the overview of various solutions based on different recommendation techniques with their advantages and disadvantages for item recommendation (see Table 1). Data mining techniques have been employed to solve the sparsity problem since they are able to analyze user buying behaviors and discover hidden relationship among items and users. Therefore, it is necessary to be adapted with data mining techniques as solutions to the sparsity problems. Moreover, it is expected to analyze the research issues

that are still unsolved or overlooked in the past studies in order to develop a new recommendation technique. Table 1 highlights research gaps in the past studies in order to motivate the current research and show the innovation of the research.

It should be noted that, based on reviews on academic research papers and issues gathered on CF research, it is obvious that even though research done in CF have obtained great development in different application fields, there are still deficiencies on music, book, joke and document recommendation systems that require further research especially with the emerging of new recommender systems applications. Therefore, in order to fill this gap, more research is required to for music, book, joke and document. MovieLens dataset has been subjected to significant research on CF, since this dataset is known as the best-known example for common users and the use of this dataset is easy. Therefore, in order to fill this gap, more researchers are required to use datasets in other application fields instead of movie. Besides, research works using practical solutions to derive user' interests from their implicit behavior is growing every year. Unfortunately, few research works have been published to grasp user' interests from social networking activities such as tagging and music listening information of the users to boost the recommendation. Hence, researchers are driven

⁸ <http://www.flixster.com>.

to develop effective techniques for dealing with such implicit data.

3. Background theories

In this section, a short introduction about important concepts of CF technique will be presented. Next, since the core of the proposed technique is based on association rules mining technique to identify the similar interest patterns between users based on users' transaction activity (like ("song played" activities), a brief explanation of this technique will be presented.

3.1. Overview of CF-based recommender systems

Basic CF systems can be divided into three steps: building user-item rating matrix, developing neighborhood foundation and rating prediction (employing the neighbors to predict unrated items and generate the recommendations) (Pirasteh, Hwang, & Jung, 2015; Ranjbar, Moradi, Azami, & Jalili et al., 2015).

CF uses the user rating data to compute degree of similarities among entities of a user-item matrix, i.e., users or items. The correlations between users and items are one of the decisive factors of predicting a user's rating for a particular item that he/she has not seen before. Thus, the central aspect in the design of recommender systems by CF is to compute the similarity between users or items with focusing on correlations between pairs of users or items. Those similar user or items are referred to as nearest neighbors and CF predicts the active user's rating on the unrated item by employing the ratings of only such neighborhood instead of whole existing user ratings. Therefore, accuracy of CF is strongly depended on selecting nearest-neighbor users successfully.

Most familiar similarity measures in CF to computing similarity degrees are Pearson's Correlation Coefficient (PCC) (see Eq. (1)) (Gogna & Majumdar, 2015) or Cosine Vector Similarity (CVS) (see Eq. (2)) (Acilar & Arslan, 2009).

$$W_{xy} = \frac{\sum_{i=1}^n (r_{xi} - \bar{r}_x)(r_{yi} - \bar{r}_y)}{\sqrt{\sum_{i=1}^n (r_{xi} - \bar{r}_x)^2 \sum_{i=1}^n (r_{yi} - \bar{r}_y)^2}} \quad (1)$$

Where n represents all items that both users x and y have rated. The notation $\bar{r}_x$ and $\bar{r}_y$ indicate the average rating of the items rated by user x and y , respectively. The similarity weight $W_{xy} \in [-1, 1]$ measures linear dependencies between two preferences/interests of users.

$$W_{xy} = \frac{\mathbf{x} \cdot \mathbf{y}}{|\mathbf{x}| \times |\mathbf{y}|} = \frac{\sum_{i=1}^n (r_{xi})(r_{yi})}{\sqrt{\sum_{i=1}^n (r_{xi})^2} \sqrt{\sum_{i=1}^n (r_{yi})^2}} \quad (2)$$

Where r_{xi} and r_{yi} are the ratings of the users x and y on item i . The notation n represents all items that both users x and y have rated.

When the similar users with active user were selected based on Eq. (1) and Eq. (2), a prediction of the active user's preferences on unrated items, is calculated. For example see Eq. (3), $pred_{xj}$ are computed as the weighted average of ratings on item j given by all users in neighborhood of active user is, as follows:

$$pred_{xj} = \bar{r}_x + \frac{\sum_{y=1}^k (r_{yj} - \bar{r}_y) \times w_{xy}}{\sum_{y=1}^k |w_{xy}|} \quad (3)$$

Where k stands for the set of users who are identified as the

neighborhood of the active user and $pred_{xj}$ is the prediction for the user u on item j .

Although these two similarity measures namely Pearson and cosine metrics (Eq. (1) and Eq. (2)) are successful at identify nearest-neighbor users, they can have challenges on working with sparse data to be resulted from: (i) large number of items that is far beyond ability of user to rate even a small fraction of them; (ii) users do not like to evaluate the viewed/purchased items. When user preferences are very sparse compared with the huge number of users and items, similarities between users or items based on Pearson and cosine metrics might be computed from only a small number of common ratings and might end up with unreliable neighborhoods to be resulted in decreasing the accuracy of the recommender systems. Fig. 1 indicates a sample of sparse user-item rating matrix in which white cells demonstrate unseen items, while black ones indicate rated items (Najafabadi & Mahrin, 2016; Kim & Kim, 2014; Shi et al., 2014).

This paper presents a technique to improve the accuracy of recommendations generated for the users in a media item recommendation system by overcoming sparsity of the data set. The core of the proposed technique is based on association rules mining technique that extends a user's profile and identify user's preferences from implicit data by extracting rules on song features including tags and song duration which users have played. This is because song dataset in MSD is so large that it is not possible to process these massive data efficiently and accurately with applying association rule mining on a normal desktop machine. Thus, process on grouping songs with different cardinalities depends on predefining the optimal number of clusters. It is based on song features.

3.2. Overview of association rules mining

The association rules mining technique has been widely adopted to improve recommendations and represent users' interests in many recommendation systems such as the ones in Tyagi and Bharadwaj (2013) and Lucas, Segreera, and Moreno (2012). The objective of this technique is to discover the interesting relationships amongst a large set of data based on describing the relationships between the typical past users' navigation paths.

Generally, users' interests can be identified by association rule mining in the form ' $A \rightarrow B$ ' (where A and B are user interests or items), which means that users interested in ' A ' are likely to be interested in ' B ' or in other words, the occurrence of item A (antecedent side of the rule) results in the occurrence of item B (consequent side of the rule) based on user transactions of a data set. For example, in marketing analysis, an association rule 'beer, water $\rightarrow$ diaper' indicates that customers who purchase beer and water are inclined to purchase diapers. Fig. 2 shows the idea of association rules on data analysis of the market basket. Rules such as "when a customer buys products X may also buy product Y with a probability of $c\%$ " or "users who liked products X , also liked products y " can be discovered (Park et al., 2012).

There are two important concepts in the association rules mining technique, support and confidence measures that rules are evaluated using such measures. The support and confidence of an association rule $X \rightarrow Y$ is defined by Eq. (4) and Eq. (5) and only the rules with having following support and confidence value is selected as the useful rules (Kardan & Ebrahimi, 2013; Tyagi & Bharadwaj, 2013):

$$\text{Support } (X \rightarrow Y) = \frac{\text{number of transactions which contain X and Y}}{\text{number of all transactions in the database}} \quad (4)$$

$$\text{Confidence } (X \rightarrow Y) = \frac{\text{number of transactions which contain X and Y}}{\text{number of transactions which contain X}} \quad (5)$$

Our motivation to apply association rules in our recommender system is based on the idea behind this technique which discovers association among items or users based on the association of data of the active user with the data of other users or on associating data of items the active user is interested in with data of other items available on the system. In fact, association rules mining describe the probability of the associations between the traceable objects (users or item) in the database. Rules generated by association rules are easy to be interpreted, and therefore, can be easily used in practice.

4. Architecture of proposed improvement technique

This section explains our proposed technique as a solution to data sparsity of the user–item matrix for making music recommendations. It is noteworthy that our technique is based on same logic with traditional CF as both uses similar behavioral patterns between users based on their overlapping interactions on items for making recommendations. But our technique in comparison to traditional CF has several advantages. Basic CF uses the users' co-rated items to compute similarity between users. When the data is sparse, determining co-rated items is difficult. Therefore, similar behavioral patterns between users cannot be computed by traditional CF and obtain poor recommendation result. While, the concept of similar preferences between users in improved technique is extended by analyzing implicit data consist of: 1) Songs played, 2) number of play counts to a specific song, 3) play ratio for certain category of songs and 4) tagging information. For example, suppose both users have rated/played different music from same genre (pop). If we use basic CF to compute the similarity between

both users, they will not be similar at all, for the reason that there is no co-played/rated music between them. However, such a result is not true in more sparse rating data. Even though both users do not have any co-played music, both of them are fans of pop music. Thus, we should consider them to be similar in case that they are sharing very similar preference on pop music. In reality, users may like to cluster items into groups, and for each group there is a corresponding cluster of users who like items in the group. Therefore, we compute the similarity between tastes of users for music based on listened group songs overlap ratio by employing clustering technique and association rules mining.



Fig. 2. Analysis of association rules on customer purchasing behavior.

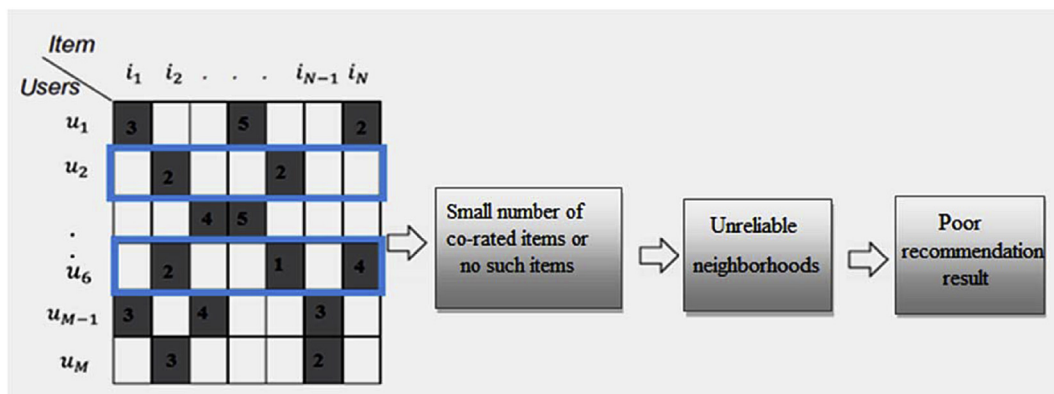


Fig. 1. Effect of sparse user-item rating matrix on recommendation accuracy in CF (Kim & Kim, 2014; Shi et al., 2014).

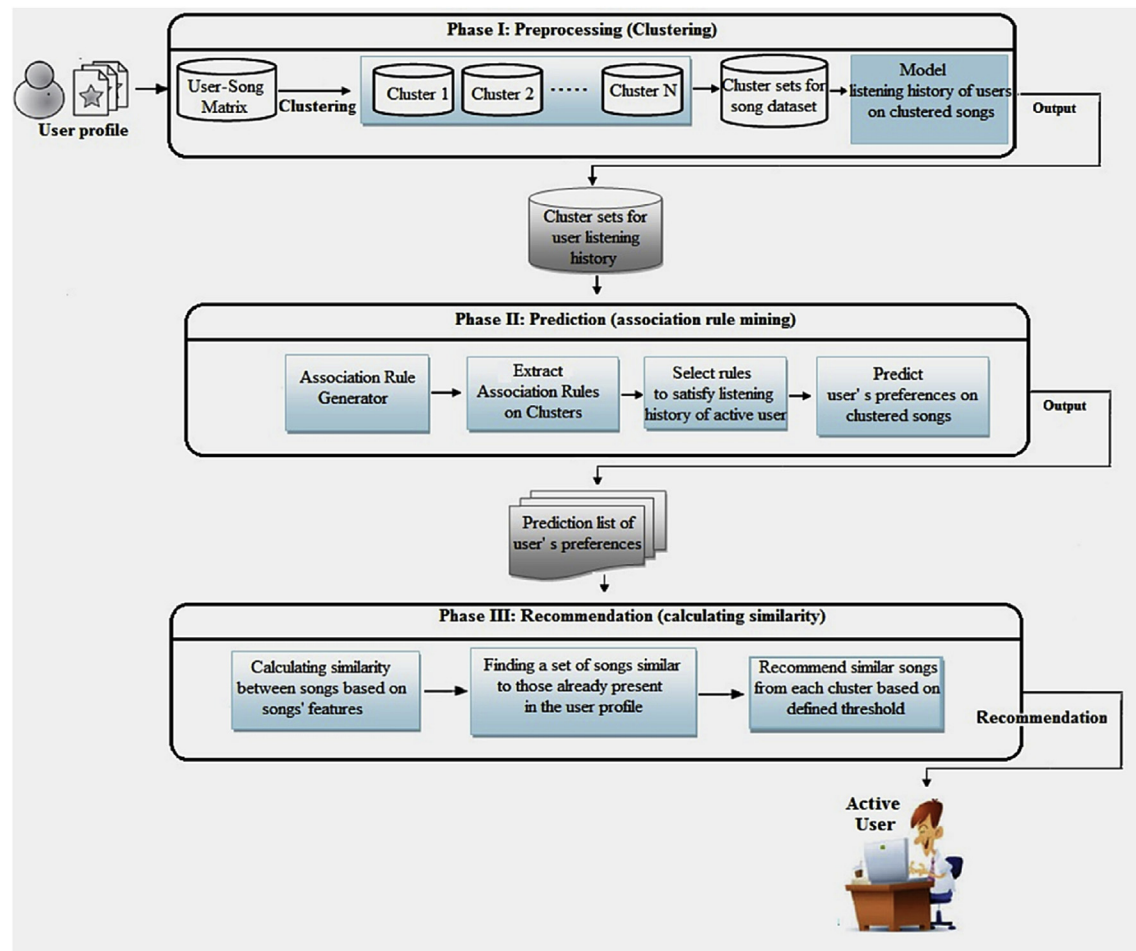


Fig. 3. System architecture of the proposed improvement technique.

Table 2

Phases and components of proposed improvement technique.

Components	Description
Phase I:Preprocessing	Reducing data dimensionality for the rule mining part
User-Song Matrix	1.Present listening history of users in form (user, song, play-count)
Cluster sets for song dataset	2.Group/Cluster songs in a hierarchical structure based on features including tags and song duration
Model listening history of users on clustered songs	3. Build a profile of a user's tastes on clustered songs based on the music he/she has played (matching song features/ cluster of tags and duration to user profiles). Express ratio of user play counts for a certain group of songs
Output: Cluster sets for user listening	Grouped songs and the profile of user's tastes on clustered songs as input for association rules mining
Phase II: Prediction	Prediction process on users' unknown preferences based on association rules mining
Association rules generator	4.Running association rules mining and generate rules according to users' listening history (user play counts on song clusters)
Extract association rules on clusters	5. Compute similarity between tastes of users on clustered songs based on their overlapping of group songs listened (Listened group songs overlap ratio). Discovering the similar listening behaviors between users (users' overlapping interactions on song clusters)
Select rules to satisfy listening history of active user	6.Find rules that their antecedent side (left side of rule) is according to user's previous preference list
Predict user' preferences on clustered songs	7. Predict user' preferences on clusters via extracting the consequent side of the rules in which antecedent side of the rules is according to user's previous preference list. Then, the consequent side of the rules is refined and just the rules with the higher preference level value between users (highest play counts) is selected.
Output: Prediction list of user' preferences	List of user' preferences on the song categories (clusters) that their score highly against the users' profiles (highest play counts)
Phase III: Recommendation	Recommend a list of songs available in each cluster that match the user' preferences
Calculating similarity between songs based on songs' features	8.Calculate the similarities between pairs of songs using the information about song's features which are the artist, year, title, release, song popularity, artist_familiarity, duration, and tags to acquire similarity degree between items
Finding a set of songs similar to those already present in the user profile	9. Compute the threshold value for specifying the similar songs with user' preferences. Then, Match the user' profile (listening history of user) with song features to recommend songs which are similar to those she or he has liked in the past.
Output: Recommend similar songs from each cluster based on threshold	10. Songs with bigger similarity value than the calculated threshold are selected for recommendation to active user.

Our technique learns an implicit correlation matrix, which links users' playing behaviors to song features by employing association rules mining to overcome sparsity issue. It works in three phases as shown at Fig. 3. Phase I is the Preprocessing Phase which do the preprocessing and song clustering. There is user profiles and user's listening behavior in the form (user, song, play_counts), where song is a music track listened by user, and play_counts show the number of times a user has listened to song. Preprocessing phase aims to form clusters or groups of songs according to their features (tags and duration of songs) and model user's preferences matrix on clustered songs. In other words, once the clusters are obtained, the cluster songs according to listening history of users are stored for extracting rules in next phase. The phase II is the prediction phase in which association rules between clusters of song features based on user's listening patterns are extracted using association rules mining. This phase predicts the users' preferences on clustered songs based on finding the similar behavioral patterns of users on their overlapping of songs played. The phase III is recommendation phase that the recommendation takes place for the active user. Here, similarity of items inside clusters or rules predicted is computed with the items already present in the user profile based on songs' features for making recommendations. The Fig. 3 shows the architecture of proposed technique with its essential components in each phase to be described in the following subsections.

Phases and components of proposed improvement technique will be described at Table 2 for easier understanding of the architecture of our technique. Then, each phases of our technique will be described as detail in the following Subsections (4.1, 4.2 and 4.3).

4.1. Preprocessing Phase

Our technique's focus is on exploiting existing information beyond the User-Item Matrix that is useful for improving recommendations. User-Item Matrix is the basis for CF technique that can be enhanced by rich side information associated with items (features such as tags, year of release, multimedia content, and etc.) and users (Job, age, gender and etc.) or interaction information about interplay of user and items (rate, purchase, click) (Najafabadi & Mahrin, 2016). In this research, we enhance User-Item Matrix by aggregating rich side information about songs from different datasets in MSD (including genre, title, duration, year of release, artist, etc.). In addition, we exploit interaction information from analyzing implicit user feedback and user's activities (songs played, number of play counts to a specific song, play ratio for certain category of songs and tagging information), rather than explicit user ratings to construct user profiles.

With the assumption that song features, especially its content features like tags are one of the main reasons for user favorite to a song, we cluster songs in a hierarchical structure based on features including tags and song duration to infer degree of user preferences on category that song belongs to. Cluster analysis technique is a research filed in data mining that partition items or users in a database into groups such that items or users belonging to same cluster have a high degree of similarity, while items or users within different clusters have a high degree of dissimilarity (Boratto et al., 2015; Huang et al., 2015). Fig. 4 shows such a hierarchical structure for MSD used in experimental methodology of the present study. More specifically, there are tags used for songs at Level 1, and each tag has 5 categories for song duration based on minutes (very short, short, medium, long, very long) at Level 2. Finally, each cluster at Level 2 shows user's listening behavior to certain category that a listened song belongs to.

It is worthwhile to say that the tags are the user actions of defining content features for songs to implicitly grasp the user interests with very important information. Firstly, social tags derivate valuable implicit sources of the genre content of each song to offer user what he/she wants without asking him/her explicitly and secondly, tags extract similar points of view on songs because understand of users about the content-based features of the songs (Horsburgh et al., 2015; Ignatov, Nikolenko, Abaev, & Poelmans, 2016; Liu, Omar, Liou, Chi, & Hsu, 2015).

Therefore, our technique involves first enriching user profiles by extracting the set of tags representative of the genre content of each song. If a song doesn't have tags, the proposed recommendation system will retrieve the genre content of each song. In MSD, there are no explicit ratings, but user's listening behavior and input data are in the form of tuples (user, song, play_count), where song represents the song listened or played by user, and play_count indicates the number of times that a user has played or listened to a particular song. In order to overcome sparsity issue and huge size of the dataset to make it more suitable for producing association rules, we utilize song features to model user's listening actions on clusters or groups of song features (see Fig. 4) to depict user preferences and detect the implicit relationships among users by applying association rule mining in the next phase (4.2.Prediction phase).

After clustering songs to certain features (tags and song duration), our proposed technique estimates the preference levels (play ratio) of a user for the cluster of songs which was played as follows:

users' preference level to a particular cluster

$$= \frac{\text{listening times to song within a particular cluster}}{\text{Maximum number of listening times in user profile}} \quad (6)$$

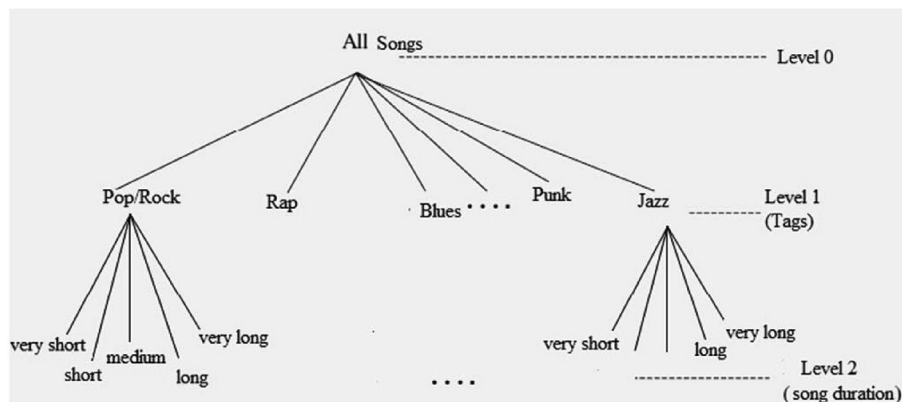


Fig. 4. Songs clustering of experimental Million Song Dataset (MSD).

Table 3
Structure of user's profile data (example).

User	Song	Play count records of songs	clustered songs	User' preference level to a particular cluster (Eq. (6))	Converted play counts to categorical variables
1	A	2	Pop/long	0.33	Medium
1	B	6	Jazz/short	1	High
2	A	5	Pop/long	0.83	High
2	D	1	Jazz/medium	0.16	Low
2	C	6	Rap/long	1	High

It is important to determine the correct number of clusters and acquire usefulness of the identified clusters as the performance of association rules mining. Therefore, tags and song duration have been selected to group songs as number of clusters and number of songs belongs to the proposed clusters are sufficient for prediction process on users' preferences based on association rules mining. Moreover, with the assumption that content features especially music tags (regarding to its genre) and song duration are the main reasons for a user preferring music and that tags used for music reflect the track music styles and music content information, we propose a clustering technique to infer the degree of users' preferences to which certain features (genre and duration) are present in any music.

It is noteworthy that association rules mining technique does not consider counting problem (item repetition within a transaction or in our case, the number of times a song was played in each user's preference list) such that it can be run on binary data (purchased or not, viewed or not, played or not) and categorical data (like sunny, rainy, overcast) (Bauer & Nanopoulos, 2014; Horsburgh et al., 2015). In other word, association rules mining was not able to generate the rules by considering the number of times a song was played by each user within a transaction. Note that a transaction in this research is said to be made regardless of a user's listening pattern. In order to use the song listening information of the users within our technique, we construct user's profile data with song' features by focusing on the amount of time that a user has listened to a particular cluster of songs based on Eq. (6) to infer the degree of the users' preferences in each song category separately. For each user, the cluster of songs with a listening time (Play count records) between the 10% and 30% percentiles are coded as "Low". The clusters related to songs that appear between the 30% and 60% percentiles of the listening time are recoded as "Medium". The songs in each cluster that the user has listened to from the 60% percentile and up are given "High" (See Table 3). Note that each user may play/listen to several songs from each cluster. In this case, the average between songs listened is taken. Forth column in Table 3 ("Clustered songs") corresponds to category that song played belongs to (by regarding the Fig. 4). Obviously, Fifth column of Table 3 demonstrates that for estimating the play ratio of a user for the cluster of songs, Eq. (6) has been applied. Last column in Table 3 indicates that how our technique deals with limitations of association rule mining technique on counting values in order to apply this technique.

It is worthwhile to mention that determining Eq. (6) and such threshold value based on user listening times was a challenging issue in order to provide appropriate equation and values for the threshold to be suitable for input data in association rule mining. To the best of our knowledge, none of previous research papers consider how to use number of times a song was played within a transaction as input for association rule mining technique.

However, our proposed technique in this research uses number of play counts to certain cluster of songs for each user profile in the form of "High", "Medium" and "Low" as input for association rule mining. The basic component of the proposed technique is association rules mining technique, which is used for identifying similar listening history among users on the same set of songs and predicting users' unknown preferences. In addition, no previous work answered the question of how to process massive data efficiently through employing association rules mining to model listening history of users. As aforementioned, this paper groups songs into clusters to deal with massive data and reduces the dimensionality of the item space and the user-item interaction matrix. The description about the application of association rule mining technique in our technique will be given in the next subsection with details.

4.2. Prediction phase

The goal of this phase is to improve prediction process in CF and overcome problem related to CVS and PCC metrics (refer Eq. (1) and Eq. (2) at section 3.1) in sparse situations where only rating data is used for computing similar preferences between users and predicting the user's interests (Bobadilla, Ortega, Hernando, & Bernal, 2012). To solve data sparsity problem, our technique involves listening history of users and extend a user's profile based on matching song features (clusters) to the user profiles, as attributes of songs could be weighted for their importance to the user. Once user profiles are obtained by tracking user listening habits, ratio of user play counts as a hint of user interest for a certain category of songs is expressed. Then, from past transactions by groups of users with similar preferences (neighbors), association rules between categories of songs are extracted using association rule mining technique. In other words, our technique predicts cluster sets of songs for active user by referencing the similar listening patterns of users based on their overlapping of sons listened.

In this research, the association rules mining analyze user profiles and user's listening behaviors to find users who share similar preferences on the same cluster of songs and then generate association rules based on played songs and user playing counts. Association rules help to discover the information about listening history of users based on similar listening activities for songs between users and then predicts active user's unknown preferences on cluster of songs. Rules generated in our proposed technique interpret two types of associations between users and items. The first type of associations to find the interesting correlation between active user's profile data with song' features. The second type shows correlation between data of active user's profile with data of other user's profiles like process of CF in recommender systems.

As aforementioned, the profile of each user is based on number of times each user has listened to particular cluster of tag and song duration. By this means, generating the association rules on user's profiles can be used to predict other cluster of tag and song duration for active user based on correlation between data of active user's profile with data of other user's profiles and correlation between active user's profile data with song' features (tag and song duration). Let us take a simple example to show this point:

"RapMedium = High" (based on active user's listening behaviors, our system denotes that songs with Rap tag and with duration of medium have been listened high times.

Association rules applied in our proposed technique generate rules with focus on seeking users (commonly called neighbors) who have similar preferences to predict their choices of songs. The basic part of generating association rules in this research is base on applied assumptions as follows:

Table 4
Samples rules extracted from listening history of users of MSD.

Rules	Description
{Rap/medium = high} $\Rightarrow$ {Blues/short = low}	Users who played songs with Rap tag and duration of medium with play ratio of high, also played songs with Blues tag and duration of short with play ratio of low
{PopRock/short = low} $\Rightarrow$ {Blues/medium = high}	Users who played songs with Pop/Rock tag and duration of short with play ratio of low, also played songs with Blues tag and duration of medium with play ratio of high

- Users who listen or play same categories of songs in past will participate in playing the similar categories of songs in future.
- Users who have the same play ratio for certain category of songs in past are likely to have the same listening behavior in future.

Based on the above assumptions, Table 4 shows sample of association rules which can be determined in this research. These association rules eventually are employed to find similar user opinions on song features based on their overlapping of songs played and predict user's new interest. To obtain a better understanding of association rules showed at Table 4, we represent the variables as shown in the example of Eq. (7) that can describe the antecedent side of the rules (left side) and consequent side of the rules (right side) as follows:

$$R1R2 = P1 \rightarrow R3R4 = P2 \quad (7)$$

- > R1, R3: clusters of tags inferred from user-song listening patterns
- > R2, R4: clusters of song duration inferred from user-song listening patterns
- > P1, P2: play ratio of users for the cluster of songs (refer to Table 3 at subsection 4.1)

Noted that, the prediction list of active user's preferences are modeled as consequent side of the rules to be generated by correlation between data of active user's profile with data of other user's profiles on the antecedent side of the rules.

In particular, the proposed technique in this research predicts cluster of songs for active users via extracting the consequent side of the rules in which antecedent side of the rules is according to active users' preference list. Then, the consequent side of the rules is refined and our system takes into account just the rules that have the higher preference level value between users ("High" score) in order to choose the cluster of songs predicted with higher a play count and thus, achieve higher prediction accuracy than the basic CF. It can be assumed that the higher a play count on song clusters, the more likeable the respective cluster is to the user (Huang et al., 2015; Xu & Yin, 2015).

Consequently, to overcome the dependency of basic CF on the users' ratings for finding similar users, our recommendation system involve association rules mining for prediction phase so that association rules in the recommender systems are able to make recommendations by using implicit information including songs played, the number of times a user has played/listened to a specific song, play ratio for certain category of songs and tagging information. It is noteworthy that Support and Confidence Value (Eq. (4) and Eq. (5)) for rules are defined and only the rules listening history of users) come with a predefined support and confidence value can be selected as the useful rules. These rules find similar user activities based on the song he or she has played. In the other words, each rule shows a cluster of songs that like-minded users prefer. When our proposed technique created the prediction list on the song categories that score highly against the users' profiles, next

phase is needed to extracts user's preferences to songs available in each cluster suggested separately.

4.3. Recommendation phase

After predicting active user's unknown preferences for a certain category of songs, this is the time to make recommendations for active user to the specific list of songs. For this phase, our technique computes the similarities between the songs using the information about song's features which are artist, year, title, release, song_hotness, artist_familiarity, duration, and tags to find a set of items similar to the items already present in the user profile. The improvement technique calculates similarities between pairs of items based on this logic that two items are similar, if they have close features to each other. The active user receives the recommendations of songs which are similar to those she or he has liked in the past. The idea of calculating the similarity degrees between songs is to match user profiles (listening history of users) with song features when generating the recommendations. The logic behind this idea is that users like to receive recommendations regarding to items (songs) which are similar to those they have selected at in the past based on song features (Ghazarian & Nematbakhsh, 2015; Horsburgh et al., 2015; Kim et al., 2011a,b; Wu, Chang, & Liu, 2014).

Recommending songs which is based on the matching song's features to the users' preference lists is a good move since song features can be weighted on their importance to the users. Thus, the importance and usefulness of song's features are investigated in predicting users' preference. The reasoning behind this according to Horsburgh et al. (2015) is that a user is likely to play a new song if its features are similar to features of songs which the user has played in the past. In order to recommend which songs from predicted clusters the user will like it was needed to compute similarity between corresponding pairs of song features as shown at Fig. 5. This figure indicates the computing similarities between pairs of songs based on their features.

It should be noted that, Similarity_degree is a number value in range [0.0, 1.0] interval as 1.0 indicates the most similar songs, while 0.0 shows the lowest similarity value between songs. The proposed technique recommends user preferences for songs according to the records of played songs by user. In fact, recommendations are made to each user based on similarity degrees between songs available in the song clusters suggested as user' preferences and the songs played by the user, as shown in Eq. (8).

$$R(u) = \frac{\sum_{s_i \in P_u} \text{Sim}(s, s_i)}{|P_u|} \quad (8)$$

where $\text{Sim}(s, s_i)$ is the computed Similarity_degree at Fig. 5, s is songs within song clusters predicted, s_i is the set of songs played by active user u . P_u is number of songs played by user.

It is noteworthy that defining a threshold is needed for specifying the similar songs in each predicted cluster from all other songs that active user has listened to. However, defining the threshold value is a challenging issue and effortful in a way that setting high threshold value causes very few songs or sometimes null song to be chosen in each cluster for recommendation purpose.

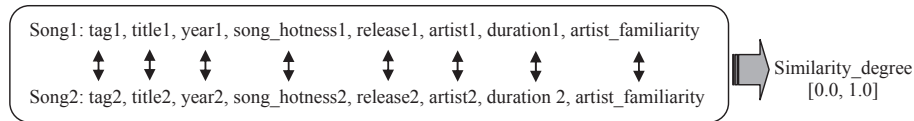


Fig. 5. The similarity degree between two songs in MSD.

On the contrary, if setting value for the threshold is very low, not very similar songs will be chosen. Thus, in order to determine the suitable threshold value, we borrow from the logic behind Boratto et al. (2015) and Huang et al. (2015) that items belong to the same cluster have a high degree of similarity, while items within different clusters have a high degree of dissimilarity. Therefore, four songs from each suggested cluster were selected (by running many experiments) to compute the average value of similarity degree between songs based on Eq. (8). This was done to generate an appropriate threshold value for specifying the similar songs with user's preferences list. It should be noted that different numbers of songs have been tested in order to determine appropriate value for the threshold and four songs were selected as the best for specifying the most similar songs from each cluster with user's preferences list. Songs with bigger similarity value than the calculated threshold are selected for recommendation to active user.

5. Experimental evaluations

This section presents the results of the experiment conducted on MSD on the improvement of the recommender's performance. The goal is to compare the behaviors regarding the proposed technique on list of recommended music among users against the basic CF under sparse data conditions. Additionally, the experiment result derived from other existing techniques on generating music recommendations for comparing with proposed technique will be presented in this section.

5.1. Evaluation dataset

Nowadays, music information retrieval technologies and music recommendation techniques have become the center of attention and they are widely used in commercially famous music recommendation systems like Last.fm, Pandora,⁹ etc. According to Park et al. (2012) and Najafabadi and Mahrin (2016), although music recommendation systems are the intense research topics in both the industry and academia, only few researches have been conducted concerning these topics. Therefore, in order to fill this gap, this paper has focused on music recommendation systems. MSD is one of the largest and free datasets in music domain. It is constructed from about one million songs and users, in which each user plays a small set of songs.

MSD was chosen for the evaluation of the proposed technique in this research because this dataset fits with our strategy on generating recommendations for different reasons: (i) To train our association rules mining model in discovering the significant correlation among the active user's profile with other user's profile. This is done using this dataset because of the need of having a dataset that has implicit information which can be extracted from users' music listening information (music played and listening times for each user) and tagging activities. (ii) To exercise our technique, a dataset with knowledge about song features (artist, year, title, release, song popularity, artist_familiarity, duration, and

Table 5

Descriptions for recommendation accuracy metrics.

$$\text{Precision} = \frac{TP}{TP + FP} \quad (9)$$

$$\text{Recall} = \frac{TP}{TP + FN} \quad (10)$$

Predicted items/actual	Relevant	Irrelevant
Recommended	True Positive (TP)	False Positive (FP)
Not recommended	False Negative (FN)	True Negative (TN)

tags) is needed to compute songs similarities. Hence, this dataset is able to provide this for us. (iii) we need a highly-sparsed dataset to show the weakness of CF, when the user-item matrix is sparse. MSD is sparse dataset, since more than 90% of the fields in user-item matrix is empty and the set of common songs in both users' preference lists becomes small or sometimes null. It is clear that due to such features of MSD that are close to the scope of this research (to present an improved CF for recommending songs), we give a particular attention to the aforementioned dataset which includes audio features about songs and their metadata.

This dataset can be linked to data in other music recommendation datasets which are musicXmatch dataset for words in lyrics, SecondHandSongs dataset for cover songs, Taste Profile subset for play count records of songs by users, and Last.fm dataset for tagging activity on songs. This research also proposes to generate user profiles from music listening history of users on tags and song duration. Furthermore, the similarity between songs based on information about song's features is used too.

5.2. Evaluation metrics

The evaluation metrics applied in this research are Precision and Recall due to their accuracy that match the goals of experiments employed in evaluating the accuracy of our proposed technique against basic CF. Precision and Recall are widely used metrics to evaluate the information retrieval researches that have been utilized by various researchers (Herlocker, Konstan, Terveen, & Riedl, 2004; Shinde & Kulkarni, 2012). The Precision metric informs us about the ability of the recommender system to indicate only relevant items among a set of irrelevant and relevant items, in contrast to the Recall metric which informs us about the recommender's capacity to recommend all relevant items. In other words, Precision computes the ratio of recommended items that are relevant and Recall computes the ratio of relevant items that are recommended. These measures are based on the number of items which are either relevant or irrelevant and either recommended or not recommended as described in Table 5. According to Table 5, the Precision and Recall accuracy metrics are calculated by Eqs. (9) and (10), respectively (Bobadilla et al., 2012; Herlocker et al., 2004):

The meaning of TP, FP and FN values are illustrated in Table 5. In this research, the total relevant items are the songs which the active

⁹ <http://www.pandora.com>.

user has showed interested in them. The precision shows how selective the system is, and the recall shows how thorough it is in selecting useful items. It is noted that the increase in the amount of items usually results in the decrease of the Precision value and an increase of the Recall value (Bobadilla et al., 2012; Herlocker et al., 2004)). Thus, another evaluation metric which is called F-measure is defined to find an optimal trade-off between Precision and Recall values. This metric is achieved by combining the Recall and Precision values within a single measure to devote an equal weight to each of them. This metric is calculated by Eq. (11) (Bobadilla et al., 2012; Shinde & Kulkarni, 2012):

$$F - \text{measure} = \frac{2 \times \text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}} \quad (11)$$

It is important to know the higher values of Precision, Recall and F-measure show the higher accuracy. We evaluate our system by considering the accuracy of Top-N recommendation verified based on Precision, Recall and F-measure.

5.3. Experimental methodology

As aforementioned, MSD documents the user's listening habits in implicit format to show what music users have listened to and how many times they do listen. Implicit format is not equivalent to CF, which uses the exact preferences of users in explicit ratings. Thus, implicit feedbacks should be converted to explicit ones in evaluating CF. In order to achieve this, we follow the logic has been employed by Langseth & Nielsen, 2015. We have focused on the number of times each user has listened to a song relative to maximum number of play counts in his profile. In such a way that songs located in the 80–100% range of user's preferences list receives a rating of 5. The songs with a listening time between the 60–80% percentiles are coded to a rating of 4. The next range of a listening time that appear between the 40–60% percentiles are mapped to a rating of 3, and so on. The aim of considering these settings is to construct the rating matrix using basic CF in order to compare the proposed technique with basic CF.

At the beginning of testing our technique, the following pre-processes were done. It was necessary to filter out all users for who with at most three play count records. This is a widely used pre-processing technique in the area of evaluating the recommender systems (Anand & Bharadwaj, 2011; Briguez et al., 2014; Horsburgh et al., 2015). In this technique, each user's listening history in the form of (user, song, play counts)-triplets is randomly split into train and test set (80% training set, 20% test set). In order to train a model in which had similarities between songs based on their existing features (like artist, title, tag, etc), these songs' features were aggregated from different datasets in MSD. In this research, users' tagging behaviors were considered. If a song did not have tag, the remaining tags were filled from the genre table in MSD. Finally, predefining of the song duration which was an integer value in seconds was done whereby it was converted to minutes.

It should be noted that in process of organizing songs into clusters based on the duration feature, a threshold value for song duration should be provided. Defining the number of clusters is the main issue for handling huge amount of data in MSD because the performance of association rules mining depends on determining the optimal number of clusters. Hence, we need to reduce the size of data for processing association rule mining and extracting association rules. Poor clustering results lead to a reduction of rules and in some cases no rules can be extracted. Thus, in order to gain the suitable value for the threshold in determining clusters based on duration of songs, different values were tested by running the experiments for categorizing songs to test the number of songs

included in each cluster and number of clusters defined as input for association rules mining.

By setting this threshold for categorization, songs with duration less than 1 min were selected for “very short” cluster. The songs that appeared between one and 3 min were coded as “short” cluster. The songs were located in cluster of “medium” with duration between three and 5 min. The songs with duration between five and 8 min were grouped in “long” cluster and the songs above the 8 min were located in “very long” cluster.

After preprocessing data in MSD, we conducted four evaluation experiments by constructing datasets with different level of sparsity to examine the accuracy of quality recommendation in our proposed recommender system against basic CF. In each evaluation experiments, users and songs were selected according to sparsity level of play count records available in selected dataset. The reason for doing experiments within different level of sparsity is that the proposed technique should be evaluated for datasets with greater variation in the sparsity levels. Hence, each dataset was randomly split into a training set which consisted 80% of preferences list per user and a testing set included the remaining preferences of user. From the training of the data to the implementation of the technique and predicting user's unknown preferences on songs, the testing set was used to evaluate the accuracy of our technique based on the relevant recommendable songs using evaluation metrics presented in the previous section.

Specifically, we verified the accuracy of our proposed technique by considering play count records with various levels of sparsity to show the impact of different sparsity levels on performance of our technique and basic CF. Various sparsity levels were achieved by creating four packages which each package contained ten groups with different sizes from user's play count records on songs. In other words, ten groups in each package were created according to classification of user's play count records. We retained 95%, 90%, 85% and so on up to 20% of the total number of training user's listening records, and discarding the rest of user's listening records to satisfy sparsity levels between “0.2–0.4”, “0.4–0.6”, “0.6–0.8” and “0.8–1.0”. It was obvious that “0.2–0.4” showed lower sparsity level and “0.8–1.0” indicated higher sparsity level. The number of users and songs in each group was based on sparsity level of the dataset to be computed as follows (Eq. (12)) (Bauer & Nanopoulos, 2014; Xie et al., 2014):

$$\text{sparsity measure} = 1 - \frac{nR}{nUsers \times nItems} \quad (12)$$

Where nR is total number of existing play count records, nUsers and nItems was the total number of users and items respectively in the User_item matrix.

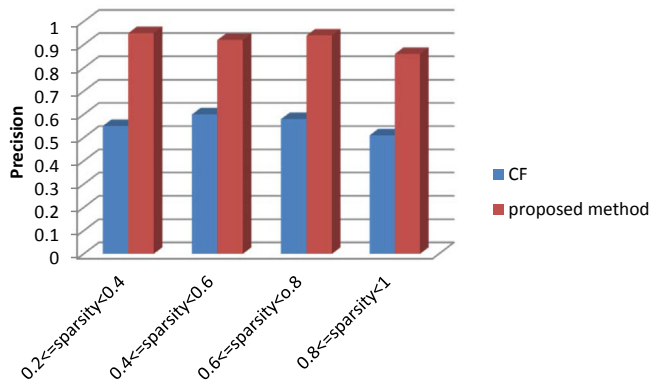
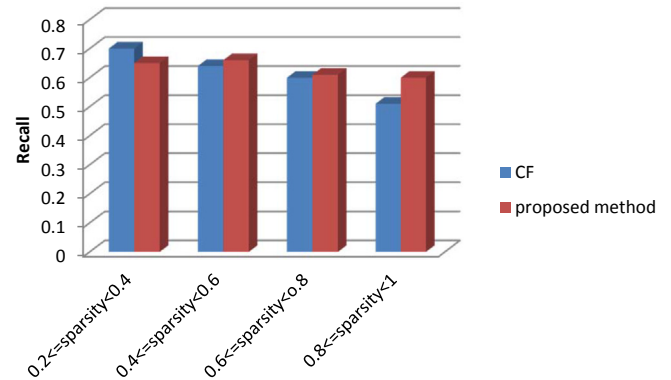
Experiments had been run on each group and then the average of experimental results of ten existing groups for each four packages was computed. The next subsection will show results obtained from conducting these four experiments. Another point to be noted was that in some cases, the number of songs played by a user, the number of users who played a particular song, and common songs in listening history of users were few or sometimes null that such cases would cause the increase in sparsity. With this consideration, the basic CF was impossible to predict user's preferences. Hence, we placed the priority with CF and used part of data for which CF could perform predictions in order to perform experiments and evaluations in sparse data. Obviously, the number of songs at lower sparsity levels cannot be high such that it could satisfy limitations of the basic CF.

The experiments were run on a machine with 4 GB of RAM and 3.1 GHz CPU. Java programming Language was used to implement the recommender systems. We used Waikato Environment for

Table 6

Average of Precision, Recall and F-measure (referred as F) values of CF and proposed technique in four experiments for MSD.

Techniques	Experiment 1 (0.2_0.4)			Experiment 2 (0.4_0.6)			Experiment 3 (0.6_0.8)			Experiment 4 (0.8_1)		
	Precision	Recall	F	Precision	Recall	F	Precision	Recall	F	Precision	Recall	F
Basic CF	0.55	0.7	0.6	0.6	0.64	0.6	0.58	0.6	0.58	0.51	0.51	0.49
Proposed technique	0.95	0.65	0.75	0.92	0.66	0.75	0.94	0.61	0.73	0.86	0.6	0.69

**Fig. 6.** Comparison of Precision metric for proposed technique and CF.**Fig. 7.** Comparison of Recall metric for proposed technique and CF.

Knowledge Analysis (WEKA)¹⁰ software tool as known library of data mining algorithms for applying association rule mining in our proposed technique.

5.4. Experimental results

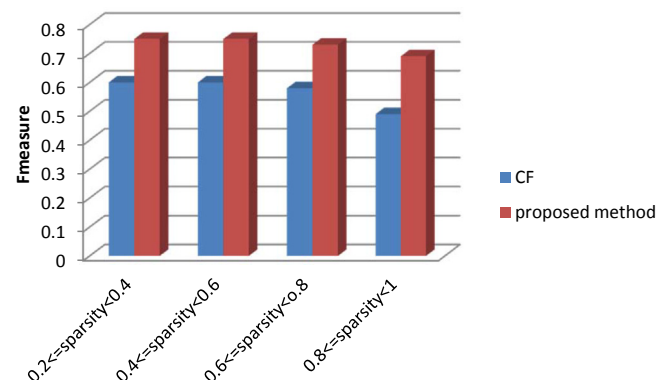
After performing the experimental settings based on the selected dataset, the results obtained from evaluating the accuracy of our technique against basic CF will be discussed. This research has employed the Precision, Recall and F-measure metrics on the relevant recommendable songs. Table 6 summarizes the accuracy of our technique and basic CF during four experiments with respect to evaluation metrics mentioned.

The results show more accurate values with respect to Precision, Recall and F-measure values for our technique in comparison to CF for all experiments except, in case of first experimental in terms of Recall. Precision, Recall and F-measure values associated to the type of experimental in CF decreases with respect to the increase in sparsity. This is expected, since higher sparsity in MSD leads to poor neighborhood formation in CF and hence provides an inaccurate representation of users' preferences.

As is clearly seen at Table 6, our technique controls the accurate values for Precision, Recall and F-measure in each four experiments, which shows its ability to manage recommendation accuracy in different sparsity measures, since our technique is not limited on user ratings to determine similarity among the users. Therefore, the experimental results show that our technique gives the better experiment results against CF at higher sparsity levels and accuracy of recommendations has been improved by over 20% in terms of F-measure metric with sparsity value more than 0.8. In general, experimental results (as illustrated in Table 6) assure that the recommender system proposed in this research will be able to recommend more accurate suggestions to the users although the data is sparse and few songs are verified by each user. This is because our technique uses implicit feedbacks (songs played,

number of play counts to a specific song, play ratio for certain category of songs and tagging information) to eliminate the limitations of basic CF which uses only the past rating records.

The advantages of the proposed technique are first; our technique exploits the existing information sources beyond the user/item matrix to enrich user profiles. In fact, our technique constructs the users' profiles based on the users' implicit data to eliminate the dependency of CF to the user's rating and enrich users' profiles based on users' preferences on tags and song duration. Second, in terms of the accuracy prediction, our proposed technique uses listening history of users who have similar played songs and similar behaviors in play count records by using association rule mining. This is because association rule mining is able to find the rules associated with the users' behaviors and hidden relations among users. An advantage of extracting association rule from user's listening behaviors based on their interests on tags and song duration is that enough information about user's interests are involved in process of finding relationships among users and predicting users' interests instead of only apply users' ratings. Third, as a matter of fact, users will probably prefer an item similar to the

**Fig. 8.** Comparison of F-measure metric for proposed technique and CF.

¹⁰ <http://www.cs.waikato.ac.nz/ml/weka/>.

ones that already existed in her/his profile, hence, similarity between songs is considered to make useful recommendations. On the other hand, the improvement that the proposed technique offers is on applying both users and item similarities based on auxiliary data and thus, the sparsity problem can be alleviated.

Fig. 6 shows the difference in Precision values between the proposed technique and basic CF among all sparsity measures increases. It is because less number of the irrelevant songs or wrong information is extracted in the proposed technique unlike basic CF. Thus, the Precision metric of our technique was compared with basic CF during four equal experimental sparsity levels in Fig. 6. It is demonstrated that our technique has higher precision value than basic CF in all sparsity measures. On the average, the precision has improved by 0.36.

Similar to the precision metric, the Recall metric is also based on relevant recommendable songs. However, it is different compared to the Precision metric in which in this metric (Recall) the number of songs which are relevant but not recommended is given ample consideration.

Fig. 7 illustrates the differences between accurate values of Recall for our proposed technique versus basic CF. It can be seen in Fig. 7 that both techniques have approximately similar behavior in sparsity levels between “0.2–0.4”, “0.4–0.6” and “0.6–0.8”. But as shown in this Fig. 7, the results from our technique present more accurate values at higher sparsity levels as our technique improves **Recall** to 9% on higher sparsity (0.8–1). It demonstrates that the proposed technique performs better than CF with respect to the Recall in situations that sparsity in user-item matrix has increased.

Fig. 8 depicts the results of comparing the satisfaction of users from recommended songs by proposed technique and basic CF with respect to the F-measure in all test cases. As mentioned in Section 5.2, the combination of the Precision and Recall values into a single metric is shown in the F-measure metric. As Fig. 8 demonstrates, by increasing values of sparsity in user-song matrix, the average satisfaction of users from recommended songs by basic CF becomes smaller while the proposed technique control the F-measure values' fluctuations in different sparsity levels and have higher F-measure values among all sparsity measures compared to basic CF. Consequently, the corresponding results to F-measure for proposed technique implemented in this study demonstrated more accurate recommendations. This improvement resulted in proposed technique due to its major advantage on achievements of sufficient information about the user's preferences to be analyzed from user's listening behaviors and eliminate its dependency on user ratings in the process of discovering users' interested items. The last three figures (Figs. 6–8) are witness of our improvements regarding Precision, Recall and F-measure.

To summarize, Figs. 6–8 have showed that in sparse situations in MSD, the accuracy of our proposed technique is improved and obtain the satisfaction of more users on recommended songs. This was done by comparing the experimental results obtained from the proposed technique with the experimental results acquired from CF basic technique. The evaluation experiments were done under similar conditions with the same datasets, in four experimental sparsity measures. The acquired experimental results validated that integrating more than one technique can avoid drawbacks of a single technique and provide better recommendations, since accuracy of generated recommendations by the proposed system against CF basic system is more significant.

5.5. Comparative evaluation with existing recommendation techniques

Findings have showed that the proposed technique substantially outperforms basic CF in terms of accuracy (see subsection 5.4). In

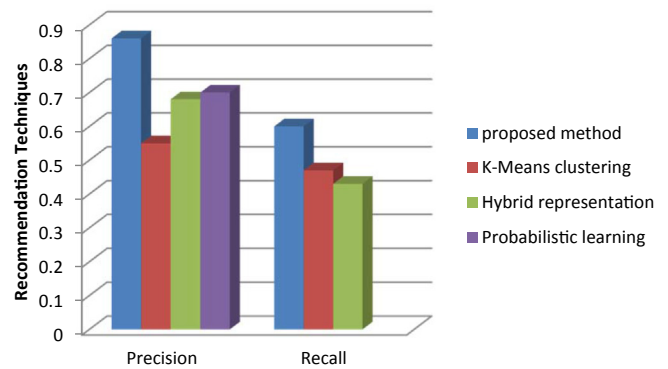


Fig. 9. Comparative evaluation of four music recommendation techniques.

this subsection, we compare the accuracy of proposed technique with other techniques which focus on music recommendations and their evaluation measures are relevant with our research work.

It is interesting to note that the evaluation of recommendation techniques in the music recommender systems in terms of accuracy metrics consist of Precision and Recall is challenging. This is because of not much research work has been conducted on this to evaluate relevant recommendable items. In terms of these accuracy metrics, recommender systems are evaluated in its accuracy and ability to retrieve the more relevant recommendable items, which maximizes users' satisfaction. Due to this reason, we compare performance of our proposed technique on recommendation accuracy with three techniques found in music recommendation research (as representative), which are techniques discussed in Zahra et al. (2015) and Horsburgh et al. (2015) and Lee et al. (2010). They have conducted experiments on the Last.fm dataset (as in our case, the Last.fm dataset is part of MSD) and give numerical results for Precision, Recall metrics:

- K-Means clustering: In Zahra et al. (2015), the authors improved recommendation process by using the k-means clustering to cluster user/item matrix based on exploiting users with maximum number of ratings. They use the binary scale from listening count and finding out the most similar centroids in k-means clustering as neighbors of active user for generating recommendations.
- Hybrid Representation: In Horsburgh et al. (2015), the authors proposed a novel approach to recommend music tracks by combining the pseudo-tags learned from similar music tracks (based on content-based retrieval) and a dynamic integration of these pseudo-tags with existing tags. Pseudo-tags improve tags on tracks that are weakly tagged based on result of tagging activity.
- Probabilistic Learning: In Lee et al. (2010), the authors solved the item sparsity problem by proposing a personalized similarity algorithm that associates the users and items with domain concepts. The proposed technique utilizes a probabilistic match with domain concepts to find similar users whereas basic CF utilizes an exact match of users' ratings to find similar users.

The experimental results presented at Fig. 9 show that our proposed technique achieves better recommendation performance than the state-of-the-art techniques on music recommendation research based on Precision, Recall metrics. As seen at Fig. 9, hybrid representation (Horsburgh et al., 2015) achieves a precision value equal to 0.68 that is greater than the precision value presented in Zahra et al. (2015), but its recall is the lowest. Because of hybrid representation's accurate in suggesting interesting music tracks.

Therefore, the number of False Positives (FPs) is low, but the number of False Negative (FNs) is high. Consequently, this results in higher precision but lower recall. Moreover, the Probabilistic learning technique in Lee et al. (2010) performs better than the hybrid representation and K-Means clustering with regard to precision achieved in Lee et al. (2010). The precision value of the proposed technique in the entire set of our experiments achieve a noticeable value (in average 0.9) to show the accuracy in suggesting interesting music tracks to users which is more important in real-world systems. Overall, the proposed technique yields better performance than the state-of-the-art methods in terms of precision and recall metrics. Because of our proposed technique has improved the effectiveness of recommendations by employing clustering technique to infer the degree of users' preferences to which certain musical features (tags and duration). In addition, we use the association rules mining technique to find similar interest patterns between users.

6. Conclusions and future work

CF recommends items to an active user based on the past rating records from like-minded users (neighbor users). Since CF cannot accurately find similar neighbors when the ratings provided by the users are very sparse, prediction accuracy in user's item preferences will be low. Hence, result in low quality recommendations. In this study, to overcome data sparsity of the user–item matrix, we have proposed a novel technique for recommendations based on user profiles created from implicit user feedback which is well suited to CF with data sparsity. The goal of this research was to improve the accuracy of recommendations by efficiently profiling user's item preferences from user's listening activities and tags associated with items and identifying similar preferences on the same category of songs. To overcome the problem of data sparsity, we have employed association rules mining technique to discover similar interest patterns among users from implicit information instead of the explicit information. In recent years it has been proved that a single algorithm is not generally able to pale the drawbacks in using basic CF and optimize recommendation accuracy. Hence, our proposed technique has improved the effectiveness of recommendations by employing clustering analysis technique to reduce the size of data and by using the association rules mining technique to find similar interest patterns between users. Moreover, the importance of item features when recommend users' preference has also been investigated. Then, the four experiments on public song dataset namely Million Song dataset was conducted to investigate accuracy of our proposed technique versus basic CF. In particular, even when the data is sparse, the proposed technique improves Precision to 36% on average of different level of sparsity, improves Recall to 9% on higher sparsity level and improves F-measure to 17% on average in comparison with basic CF. The experimental results also demonstrated that our technique achieves the better performance when compared to other extended version of CF techniques in terms of Precision, Recall metrics. This improvement resulted in proposed technique due to its major advantage on achievements of sufficient information about the user's preferences to be analyzed from user's listening behaviors and eliminate its dependency on user ratings in the process of discovering users' interested items.

Incorporating item repetition within a transaction (each user buys/uses a set of items frequently) into our proposed technique provides a significant implication for research when no previous work had been found to employ users' implicit interaction records with items to process massive data efficiently through employing association rules mining. So, this research proposed a new way to use item repetition within a transaction which is well suited to association rules mining and can achieve high recommendation

accuracy. This research is first academic research to solve sparsity problem in CF technique by using the correlations between items and users in huge size of dataset with the application of the clustering and association rules mining technique. Our recommendation technique uses the user activity data logs in the form (user, clustered songs, play counts) where clustered songs are songs played by user and play counts to certain cluster of songs for each user's profile are represented in the form of "High", "Medium" and "Low" as input for association rules mining. In addition, this research employs the clustering technique for efficiently profiling user's item preferences from play records of songs and as a data dimensionality reduction for the rule mining part. However, this research can be extended to provide our research method for other recommender system examples as well such as movies or books since, association rules mining can alleviate the sparsity weakness. Because association rules do not need to consider all opinions from users in order to build a recommender model. Such technique can employ information acquired from transactions of groups of users with similar preferences (neighbors) in order to discover association rules about Web objects.

As future work, we intent to utilize more implicit users' feedback in the process of association rules mining to generate recommendations by identify similar users' activities. Hereby, the accuracy and quality of the provided recommendation will be improved by employing more implicit data extracted from association rules. Furthermore, we are planning to peruse additional data sources like users' demographic data to combine them with implicit data for extracting more reliable and more accurate rules based on useful information about the users and their interaction with items. The experiments conducted involve one dataset named MSD that contains implicit data on user's listening history and can be linked to other sibling datasets (music recommendation datasets) to provide more information on listening history of the users and the songs. Moreover, it is also suggested to continue the experimental evaluations in other fields like images, books and TV programs.

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